Hana Fujimoto

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Machine Learning Engineer (formerly civil engineering)

SKILLS

python, pytorch, onnx, opencv, cuda, numpy, ray, mlflow, polars

EDUCATION

MEng Civil Engineering, Kyoto University Apr 2012 – Mar 2014

Certificate in Applied Statistics, Osaka Institute of Technology Apr 2021 – Dec 2021

BEng Civil Engineering, Kyoto University Apr 2008 – Mar 2012

CERTIFICATIONS

Professional Engineer, Civil (Japan) — JSCE, Mar 2018

Deep Learning Specialisation — Coursera, Sep 2021

JDLA Deep Learning for Engineers — JDLA, Jul 2022

EXPERIENCE

Machine Learning Engineer

Sakura Vision

Osaka, JapanApr 2023 – Present

- Ships defect-detection models to 40 factory edge boxes acros seven client sites in Kansai and Chubu.
- Cut inference cost 60% by exporting to ONNX and quantising INT8, with a calibration set per line.
- Owns the retraining loop: Ray for sweeps, MLflow for the registry, and a shadow-deploy gate before promotion.
- Rewrote the data layer on Polars; nightly aggregation went fr 50 minutes to under four.
- Wrote the drift monitor that catches lens fouling before the fal positive rate reaches the operators.

Computer Vision Engineer

Rinku Robotics

Osaka, JapanJan 2022 – Mar 2023

- Built the bin-picking pose estimator in PyTorch.

Machine Learning Intern

Naniwa Data Lab

Osaka, JapanJul 2021 – Dec 2021

- Six months part-time alongside the bridge role, which is how transition was funded.
- Built the baseline segmentation model later open-sourced as bridge crack dataset.
- Presented weekly to a reading group of nine engineers on papers from CVPR and MVA.

Structural Engineer

Kansai Bridge & Structure

Kobe, JapanApr 2014 – Dec 2021

- Load assessment for 60 prefectural road bridges under the 2 revision of the Specifications for Highway Bridges.
- Led the seismic retrofit design for the Muko River crossing, a 340m continuous steel box girder.
- Ran finite-element models in Abaqus for fatigue life on orthotropic decks; published the calibration set.
- Supervised four junior engineers and the drawing office throu two prefectural audits.
- Wrote the Python tooling that started the career change: a cra width measurement pipeline over inspection photos that replaced two days of manual tracing per bridge.
- Standardised the inspection photo archive, 240k images, whi later became the training set at Rinku.

- Represented the firm on the prefectural committee for bridge asset management, 2019-2021.

PROJECTS

Bridge crack segmentation pytorch, opencv
Open dataset and baseline; 1.1k GitHub stars.

PUBLICATIONS

Crack width estimation on orthotropic steel decks from inspection photography. Journal of Bridge Engineering, Jun 2021

Transfer learning from civil inspection imagery to industrial defect detection. MVA, Jul 2023

A calibration set for INT8 export of small segmentation models. worksh paper, Nov 2024